

Development test Report

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Classification of Evaluate Item			Outline of Evaluate Condition and Principal Standard Value			Test result		
Test Item	Detail	Meas. item num	Depend on	MCP	Judg			
Cooling capacity Erp A (Rate)	ISO 5151							
	JIS B8815-2:1999, EN14511 In DB: 27.0 °C, WB: 19.0 °C Out DB: 35.0 °C, WB: 24.0 °C	1 (1)	Spec.: 1φ, 50Hz, 230V Q = 12,000 kW P = 3,600 kW EER = 3.333 WW	Target in Development (Risk): Q: ≥ 100.0 % P: - EER: ≥ 100.0 %	A	OK		
	ISO 5151							
Cooling capacity Erp B	JIS B8815-2:1999, EN14511 In DB: 27.0 °C, WB: 19.0 °C Out DB: 30.0 °C, WB: 20.0 °C	1 (2)	Spec.: 1φ, 50Hz, 230V Q = 8,800 kW P = 1,750 kW EER = 5.057 WW	Target in Development (Risk): Q: ≥ 100.0 % P: - EER: ≥ 100.0 %	A	OK		
	ISO 5151							
	JIS B8815-2:1999, EN14511 In DB: 27.0 °C, WB: 19.0 °C Out DB: 25.0 °C, WB: 16.0 °C	1 (3)	Spec.: 1φ, 50Hz, 230V Q = 5,800 kW P = 0,650 kW EER = 8.923 WW	Target in Development (Risk): Q: ≥ 100.0 % P: - EER: ≥ 100.0 %	A	OK		
Cooling capacity Erp C	ISO 5151							
	JIS B8815-2:1999, EN14511 In DB: 27.0 °C, WB: 19.0 °C Out DB: 20.0 °C, WB: 12.0 °C	1 (4)	Spec.: 1φ, 50Hz, 230V Q = 5,400 kW P = 0,470 kW EER = 11.489 WW	Target in Development (Risk): Q: ≥ 100.0 % P: - EER: ≥ 100.0 %	A	OK		
	ISO 5151							
Cooling capacity POFF	JIS B8815-2:1999, EN14511 In DB: 27.0 °C, WB: 19.0 °C Out DB: 20.0 °C, WB: 12.0 °C	1 (5)	Spec.: 1φ, 50Hz, 230V Q = 5,400 kW P = 0,470 kW EER = 11.489 WW	Target in Development (Risk): Q: ≥ 100.0 % P: - EER: ≥ 100.0 %	A	OK		
	ISO 5151							
	JIS B8815-2:1999, EN14511 In DB: 27.0 °C, WB: 19.0 °C Out DB: 20.0 °C, WB: 12.0 °C	1 (6)	Spec.: 1φ, 50Hz, 230V Q = 5,400 kW P = 0,470 kW EER = 11.489 WW	Target in Development (Risk): Q: ≥ 100.0 % P: - EER: ≥ 100.0 %	A	OK		
Cooling capacity PTO*	ISO 5151							
	JIS B8815-2:1999, EN14511 In DB: 27.0 °C, WB: 19.0 °C Out DB: 20.0 °C, WB: 12.0 °C	1 (7)	Spec.: 1φ, 50Hz, 230V Q = 5,400 kW P = 0,470 kW EER = 11.489 WW	Target in Development (Risk): Q: ≥ 100.0 % P: - EER: ≥ 100.0 %	A	OK		
	ISO 5151							
Heating capacity TOL	JIS B8815-2:1999, EN14511 In DB: 20.0 °C, WB: 15.0 °C Out DB: -15.0 °C, WB: - °C	2 (1)	Spec.: 1φ, 50Hz, 230V Q = 5,700 W P = 3,000 W COP = 1.907 WW	Target in Development (Risk): Q: ≥ 100.0 % P: - COP: ≥ 100.0 %	A	OK		
	ISO 5151							
	JIS B8815-2:1999, EN14511 In DB: 20.0 °C, WB: 15.0 °C Out DB: -15.0 °C, WB: - °C	2 (2)	Spec.: 1φ, 50Hz, 230V Q = 5,700 W P = 3,000 W COP = 1.907 WW	Target in Development (Risk): Q: ≥ 100.0 % P: - COP: ≥ 100.0 %	A	OK		
Heating capacity Erp B	ISO 5151							
	JIS B8815-2:1999, EN14511 In DB: 20.0 °C, WB: 15.0 °C Out DB: 2.0 °C, WB: 1.0 °C	2 (3)	Spec.: 1φ, 50Hz, 230V Q = 4,400 W P = 1,030 W COP = 4.272 WW	Target in Development (Risk): Q: ≥ 100.0 % P: - COP: ≥ 100.0 %	A	OK		
	ISO 5151							
Heating capacity Erp C	JIS B8815-2:1999, EN14511 In DB: 20.0 °C, WB: 15.0 °C Out DB: 7.0 °C, WB: 6.0 °C	2 (4)	Spec.: 1φ, 50Hz, 230V Q = 4,300 W P = 0,780 W COP = 5.590 WW	Target in Development (Risk): Q: ≥ 100.0 % P: - COP: ≥ 100.0 %	A	OK		
	ISO 5151							
	JIS B8815-2:1999, EN14511 In DB: 20.0 °C, WB: 15.0 °C Out DB: 12.0 °C, WB: 11.0 °C	2 (5)	Spec.: 1φ, 50Hz, 230V Q = 3,500 W P = 0,590 W COP = 5.932 WW	Target in Development (Risk): Q: ≥ 100.0 % P: - COP: ≥ 100.0 %	A	OK		
Cooling Std. (System A)	In 27.0 / 19.0 °C, Out 35.0 / 24.0 °C (C-1)	3 (1)	The compressor is operated in the constant Low noise mode and SHD > 10 deg	-	1	OK		
	In 27.0 / 19.0 °C, Out 35.0 / 24.0 °C (C-2)	3 (2)	The compressor is operated in the constant Low noise mode and SHD > 10 deg	-	1	OK		
	In 27.0 / 23.0 °C, Out 46.0 / 26.0 °C (C-3)	4 (1)	The compressor is operated in the constant Low noise mode and SHD > 10 deg	-	1	OK		
Cooling overcool Operating (System A)	In 32.0 / 23.0 °C, Out 46.0 / 26.0 °C (C-4)	4 (2)	The compressor is operated in the constant Low noise mode and SHD > 10 deg	-	1	OK		
	In 20.0 / - °C, Out 7.0 / 6.0 °C (H-1)	5 (1)	The compressor is operated in the constant Low noise mode and SHD > 10 deg	-	1	OK		
	In 20.0 / - °C, Out 7.0 / 6.0 °C (H-2)	5 (2)	The compressor is operated in the constant Low noise mode and SHD > 10 deg	-	1	OK		
Heating Std. (System A)	In 20.0 / - °C, Out 7.0 / 6.0 °C (H-3)	5 (3)	The compressor is operated in the constant Low noise mode and SHD > 10 deg	-	1	OK		
	In 20.0 / 15.0 °C, Out 2.0 / 1.0 °C (H-4)	6 (1)	The compressor is operated in the constant Low noise mode and SHD > 10 deg	-	1	OK		
	In 20.0 / 15.0 °C, Out 2.0 / 1.0 °C (H-5)	6 (2)	The compressor is operated in the constant Low noise mode and SHD > 10 deg	-	1	OK		
Heating low temp (System A)	In 20.0 / 15.0 °C, Out 2.0 / 1.0 °C (H-6)	7 (1)	The compressor is operated in the constant Low noise mode and SHD > 10 deg	-	1	OK		
	In 27.0 / 24.0 °C, Out 22.0 / - °C (C-1)	8 (1)	No dew formation on vane and jumping from the indoor unit	-	1	OK		
	In 27.0 / 24.0 °C, Out 22.0 / - °C (C-2)	8 (2)	No dew formation on vane and jumping from the indoor unit	-	1	OK		
Heating very low ambient temp (System C)	In 20.0 / 15.0 °C, Out -35.0 / - °C (H-3)	9 (1)	The compressor is operated in the constant Low noise mode and SHD > 10 deg	-	1	OK		
	PWL (Cool / Heat mode)	10 (1)	Outdoor - dBA Not less than -30dBA, no abnormal noise	-	-	-		
	SPL (JS condition)	10 (2)	Heat mode Outdoor 48.0 dBA Not over than 0.0 dBA Fan speed outdoor 480 rpm	-	-	-		
Noise	Sound for sweep to compressor frequency	10 (3)	Heat mode Outdoor 48.0 dBA Not over than 0.0 dBA Fan speed outdoor 510 rpm	-	-	-		
	Feeling check is OK, Then Judge is OK	10 (4)	Feeling check is OK, Then Judge is OK	-	-	-		
	Feeling check is NG, Then Judge is NG	10 (5)	Feeling check is NG, Then Judge is NG	-	-	-		
Vibration	OA-Maxpeak ≥ 6 dB(A)	11 (1)	(Cooling Mode) Vibration Peak amplitude less than 200 μm (Fan Mode) less than 350 μm	-	-	-		
	Guards 500 μm (Peak amplitude) Main body firing part = 100 μm (RMS)	11 (2)	(Cooling Mode) Vibration half amplitude less than 150 μm	-	-	-		
	Piping Vibration	12 (1)	(Cooling Mode / Heating Mode) Vibration half amplitude less than 350 μm	-	-	-		
Piping Stress [Cooling Mode]	Effect parts acceleration	13 (1)	(Cooling Mode / Heating Mode) Vibration half amplitude less than 350 μm	-	-	-		
	Start - Stop	14 (1)	(Cooling Mode / Heating Mode) Vibration half amplitude less than 350 μm	-	-	-		
	Operate	14 (2)	Piping Stress less than 36.7 N/mm2	-	-	-		
Piping Stress [Heating Mode]	Start - Stop	15 (1)	Piping Stress less than 26.8 N/mm2	-	-	-		
	Operate	15 (2)	Piping Stress less than 36.7 N/mm2	-	-	-		
	Close valve	16 (1)	Piping Stress less than 26.8 N/mm2	-	-	-		
Refrigerant leakage	Refrigerant amount 0.25 ~ 50.70%	17 (1)	It's necessary to be able protect it, TC1.2: Coil ≤ 130 °	-	-	-		
	Conductivity emission	18 (1)	Power supply terminal ≤ 56 dBμV / Terminal connecting to indoor ≤ 74 dBμV	-	-	-		
	Power emission	18 (2)	Safety SPEC by TOKIN Test	-	-	-		
Current leakage	Noise endurance (Impulse / burst)	19 (1)	2 mA or less per 1 kW (MAX 10 mA or less)	-	-	-		
	Automatic noise	19 (2)	Safety SPEC by 2.4 kV / 3.5 kV Input	-	-	-		
	Thunder surge	19 (3)	No misoperation during 10,000 ON/OFF	-	-	-		
Voltage Dip Immunity Short Interruptions Immunity	SW debug (elect board only)	20 (1)	Safety SPEC by 0.0 kV / 0.0 kV Input	-	-	-		
	SW debug (unit)	20 (2)	There is no reckless run in 30 ms. There is no reckless run in 500 ms.	-	-	-		
	SW debug (unit)	20 (3)	Operation according to SW specifications	-	-	-		
Electric board test	Life of each parts	21 (1)	Operation according to SW specifications	-	-	-		
	Electric board test	21 (2)	Calculation results of capacitor life ≥ 15 Years	-	-	-		
	Heat cycle ≥ 200 Hr	21 (3)	Heat cycle ≥ 200 Hr	-	-	-		
(Remarks) Amount of Unit			o : Refer Base model			x : not require for evaluation		
Main Problem			Δ : Pass by Have Condition			x : Not Pass		

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